

CHAPTER- VIII

JAGGED ARRAY:

- Jagged array is a two dimensional array where each row can have a different number of columns.

EXAMPLE(USER INPUT):

```
import java.util.*;

public class MarksJag {

    public static void main(String [] args) {

        Scanner sc= new Scanner(System.in);

        int[][] marks= new int[3][];

        marks[0]= new int[2];
        marks[1]= new int[4];
        marks[2]=new int[3];

        System.out.println("Enter the Marks of the Students:");

        for(int i=0; i<marks.length;i++) {

            for(int j=0; j<marks[i].length;j++) {

                marks[i][j]=sc.nextInt();

            }

        }

        System.out.println("Marks of the Students");

        for(int i=0; i<marks.length; i++) {

            for(int j=0; j<marks[i].length;j++) {

                System.out.print(marks[i][j]+" ");

            }

        }

    }

}
```

```
        }  
        System.out.println();  
    }  
}
```

ADVANTAGES:

- Saves Memory
- Flexible Structure
- Better Representation
- Faster Access

STRING IN JAVA:

- A String is a sequence of character enclosed within double quotes(" ").
- String is immutable.
- Example: String name = "Abi";

EXAMPLE:

```
public class StringExample {  
    public static void main(String[] args) {  
  
        String name = "Shalini";  
  
        System.out.println(name);  
    }  
}
```

IMPORTANT METHODS OF STRING CLASS:

- `length()`:
 - Returns the number of characters.
 - `System.out.println("Hello".length());`

- `charAt()`:
 - Returns character at index.
 - `char c= "Hello".charAt(2);`

- `substring()`:
 - Extract part of the string.
 - `System.out.println("Java".substring(2));`

- `equals()/equalsIgnoreCase()`:
 - `System.out.println("Java".equals("Java")); // true`
 - `System.out.println("Java".equalsIgnoreCase("JAVA"))`
; // true

- `toUpperCase()/toLowerCase()`:
 - `System.out.println("java".toUpperCase()); // JAVA`
 - `System.out.println("JAVA".toLowerCase()); // java`

- `replace()`:
 - `System.out.println("Java".replace('a', 'o')); // Jovo`

- `split()`:
 - `System.out.println("Java Python C".split(" ")[1]); //`
Python

➤ trim():

- System.out.println(" Java ".trim()); // Java

EXAMPLE:

```
public class StringMethods {  
    public static void main(String[] args) {  
  
        String str = "Java Programming";  
  
        System.out.println("Length: " + str.length());  
        System.out.println("Upper: " + str.toUpperCase());  
        System.out.println("Lower: " + str.toLowerCase());  
        System.out.println("Substring: " + str.substring(5));  
        System.out.println("Equals: " + str.equals("Java Programming"));  
        System.out.println("Equals Ignore Case: " +  
"Java".equalsIgnoreCase("JAVA"));  
        System.out.println("Replace: " + str.replace('a', 'o'));  
  
        String[] words = str.split(" ");  
        System.out.println("Split: " + words[1]);  
  
        System.out.println("Trim: " + " Java ".trim());  
    }  
}
```

